

Education

PhD	Bio-informatics , Eliance/INRAE/Paris Saclay Deep Learning, Quantitative Genetics, Genomic Prediction, Statistical models, Data Simulation, Animal Breeding	Jouy en Josas, France 2022 – 25
Master 2	Bio-informatics , Paris Cité University Relevant courses: Programming and Project Management, Artificial Intelligence and Optimization, Omics Applications, Structural bioinformatics.	Paris, France 2021 – 22
Master 1	STA2E- Integrative Biology (OMICS) , Paris-Est Créteil University Créteil Relevant courses: Bioinformatics, OMICS platforms, Biostatistics.	Créteil, France 2020 – 21
Bachelor	Biology , Lebanese University Relevant courses: Biomolecular, Biotechnology, Genetics, Biostatistics.	Lebanon 2017 – 20

Research Internships

Deep Learning and Genomics , Roslin Institute – Highlander Lab - Design mating plans for multiple traits, both linearly and non-linearly related, using Deep Learning to enhance genetic gain and decrease computational time. Advisor: Gregor Gorjanc	Edinburgh, Scotland March – May 2024
Deep Learning and Genomics , INRAE-GABI - Deep Learning-based model to predict phenotype using single-nucleotide polymorphism data (Python/Pytorch), application of statistical methods for Genomic prediction, GWAS. Advisors: Pascal Croiseau and Beatriz Cuyabano	Jouy en Josas, France January-July 2022
Single-cell RNA seq analysis , Inserm TAGC-U1090 - Single-cell RNA-seq analysis with R to detect PTK7 expression in subcutaneous lymph nodes and tumor (melanoma). Advisor: Aitor Gonzalez	Marseille, France April-May 2021

Publications

- Shokor, F.**, et al. (2025). Deep Learning for genomic prediction accounting for heterosis in crossbreeding systems. bioRxiv <https://doi.org/10.1101/2025.07.24.666530>
- Shokor, F.**, et al. (2025). Deep learning and GBLUP integration: An approach to identify potential nonlinear genetic relationships between traits. Journal of Dairy Science. <https://doi.org/10.3168/jds.2024-26057>

Scientific communications

- Oral presentation:** Shokor et al. (2025) Deep Learning for Crossbred Genomic Prediction: Integrating Transfer Learning and Heterosis Effects. 76th Annual Meeting of the European Association for Animal Production.
- Poster presentation:** Shokor et al. (2024) GBLUP and Deep Learning Integration: A Novel Approach to Evaluating Non-linearly Related Genetic Traits. 7th International Conference of Quantitative Genetics.
- Oral presentation:** Shokor et al. (2023) Exploring non-linear genetic relationships between correlated traits. 74th Annual Meeting of the European Association for Animal Production.

Skills and languages

Programming & AI: Python, R, Bash, Git, Jupyter, Scanpy, Seurat, Pytorch, Machine Learning, Fine-tuning, Transfer Learning
Bioinformatics: Quantitative genetics, GWAS, Single-cell RNA/CITE/ATAC-seq, Spatial omics, Multi-omics Intergation, Nextflow
Languages: English, French, Arabic

Referees

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